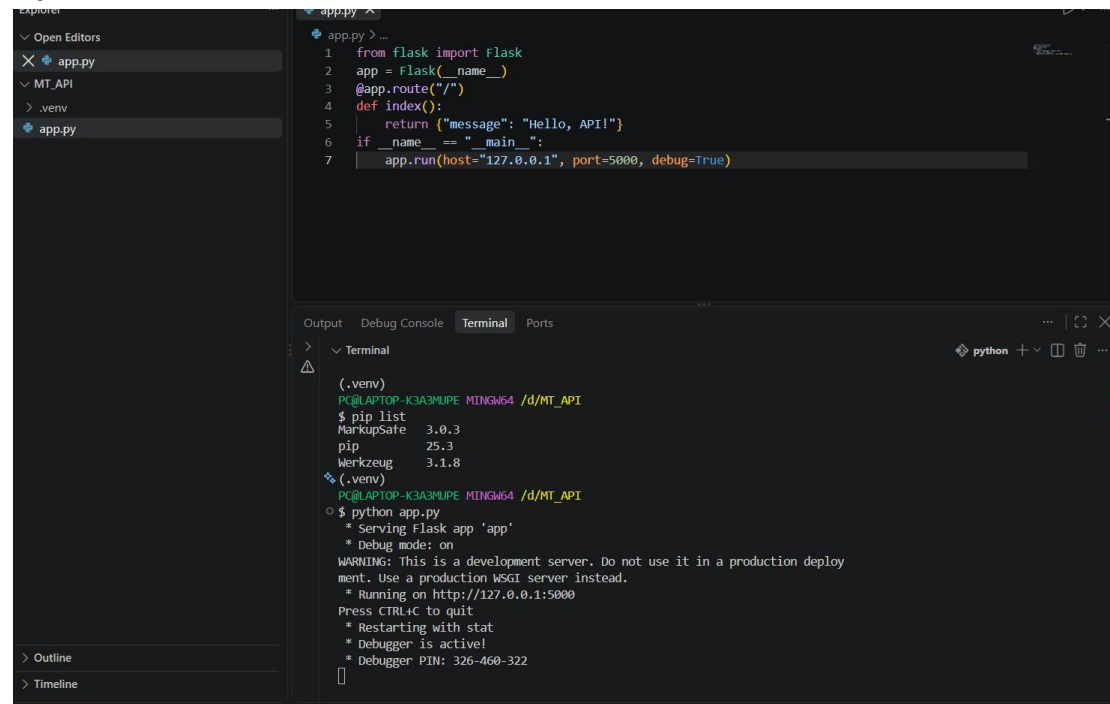


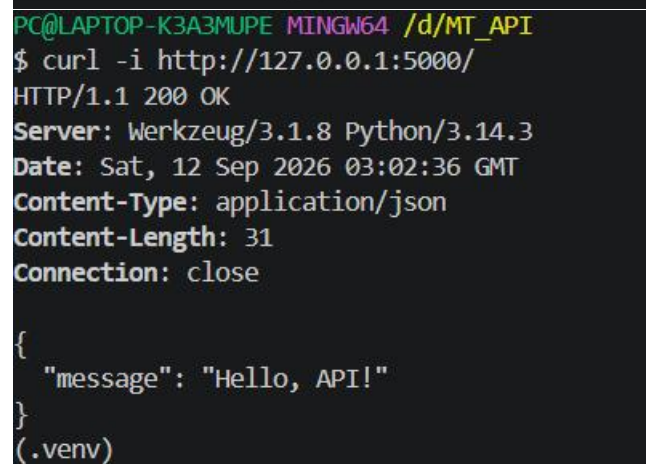
Bài 1:



The screenshot shows the Visual Studio Code editor with a file explorer on the left containing 'app.py'. The main editor displays the code for 'app.py', which is a simple Flask application. Below the editor, the 'Terminal' panel is open, showing the command prompt and the execution of the application.

```
1 from flask import Flask
2 app = Flask(__name__)
3 @app.route("/")
4 def index():
5     return {"message": "Hello, API!"}
6 if __name__ == "__main__":
7     app.run(host="127.0.0.1", port=5000, debug=True)
```

```
(.venv)
PC@LAPTOP-K3A3MUPE MINGW64 /d/MT_API
$ pip list
MarkupSafe 3.0.3
pip 25.3
Werkzeug 3.1.8
(.venv)
PC@LAPTOP-K3A3MUPE MINGW64 /d/MT_API
$ python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 326-460-322
```



The screenshot shows a terminal window with the output of a curl command. The output displays the HTTP status, headers, and the JSON response from the application.

```
PC@LAPTOP-K3A3MUPE MINGW64 /d/MT_API
$ curl -i http://127.0.0.1:5000/
HTTP/1.1 200 OK
Server: Werkzeug/3.1.8 Python/3.14.3
Date: Sat, 12 Sep 2026 03:02:36 GMT
Content-Type: application/json
Content-Length: 31
Connection: close

{
  "message": "Hello, API!"
}
(.venv)
```

Bài 2:

```
app.py X
app.py > ...
11 @app.route("/echo", methods=["POST"])
12 def echo():
13     data = request.get_json(silent=True) or {}
14     return jsonify({"you_sent": data}), 200
15
16 if __name__ == "__main__":
17     app.run(host="127.0.0.1", port=5000, debug=True)
```

Output Debug Console Terminal Ports

> Terminal

```
PC@LAPTOP-K3A3MUPE MINGW64 /d/MT_API
$ source /d/MT_API/.venv/Scripts/activate
(.venv)
PC@LAPTOP-K3A3MUPE MINGW64 /d/MT_API
$ python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 326-460-322
```

```
PC@LAPTOP-K3A3MUPE MINGW64 /d/MT_API
$ curl -i http://127.0.0.1:5000/health
HTTP/1.1 200 OK
Server: Werkzeug/3.1.8 Python/3.14.3
Date: Sat, 12 Sep 2026 03:30:09 GMT
Content-Type: application/json
Content-Length: 21
Connection: close

{
  "status": "ok"
}
(.venv)
PC@LAPTOP-K3A3MUPE MINGW64 /d/MT_API
$ curl -X POST http://127.0.0.1:5000/echo -H "Content-Type: application/json" -d '{"name":"An", "age":21}'
{
  "you_sent": {
    "age": 21,
    "name": "An"
  }
}
```

Bài 3:

```
app.py X
app.py > ...
8 def create_student():
12     return jsonify({"error": "name là bắt buộc"}), 400
13
14     student = {
15         "id": str(uuid4()),
16         "name": name,
17         "gpa": body.get("gpa", 0.0),
18     }
19     STUDENTS.append(student)
20     return jsonify(student), 201
21
22 if __name__ == "__main__":
23     app.run(host="127.0.0.1", port=5000, debug=True)

Output Debug Console Terminal Ports
>
Terminal
PC@LAPTOP-K3A3MUPE MINGW64 /d/MT_API
$ source /d/MT_API/.venv/Scripts/activate
(.venv)
PC@LAPTOP-K3A3MUPE MINGW64 /d/MT_API
$ python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 326-460-322

PC@LAPTOP-K3A3MUPE MINGW64 /d/MT_API
$ curl -i -X POST http://127.0.0.1:5000/students -H "Content-Type: application/json" -d '{"name": "An", "gpa": 3.4}'
HTTP/1.1 201 CREATED
Server: Werkzeug/3.1.8 Python/3.14.3
Date: Sat, 12 Sep 2026 03:32:40 GMT
Content-Type: application/json
Content-Length: 81
Connection: close

{
  "gpa": 3.4,
  "id": "3472bd81-b868-4413-ab19-6e56e99d80b9",
  "name": "An"
}
(.venv)
PC@LAPTOP-K3A3MUPE MINGW64 /d/MT_API
$ curl -i -X POST http://127.0.0.1:5000/students -H "Content-Type: application/json" -d '{"gpa": 3.4}'
HTTP/1.1 400 BAD REQUEST
Server: Werkzeug/3.1.8 Python/3.14.3
Date: Sat, 12 Sep 2026 03:33:43 GMT
Content-Type: application/json
Content-Length: 49
Connection: close

{
  "error": "name l\u00e0 b\u1eaft bu\u1ed9c"
}
(.venv)
```

Bài 4:

```
app.py X
app.py > ...
15 def get_book(book_id):
16     if book is None:
17         return jsonify({"error": "not found"}), 404
18     return jsonify(book), 200
19
20
21 # Path param: Ép kiểu int ngay từ URL
22 @app.route("/items/<int:item_id>")
23 def get_item(item_id):
24     return jsonify({"id": item_id}), 200
25
26 # Query string: Lọc và phân trang
27 @app.route("/books", methods=["GET"])
28 def list_books():
29     limit = int(request.args.get("limit", 20))
30     q = request.args.get("q", "").strip().lower()
31     items = [b for b in BOOKS if q in b["t"].lower()]
32     return jsonify({"items": items}), 200
33
34 if __name__ == "__main__":
35     app.run(host="127.0.0.1", port=5000, debug=True)
```

Output Debug Console Terminal Ports

> Terminal

- (.venv)
PC@LAPTOP-K3A3MUPE MINGW64 /d/MT_API
 - o \$ python app.py
 - * Serving Flask app 'app'
 - * Debug mode: on
 - WARNING: This is a development server. Do not use it in a production deployment. Use a product on WSGI server instead.
 - * Running on http://127.0.0.1:5000
 - Press CTRL+C to quit
 - * Restarting with stat
 - * Debugger is active!
 - * Debugger PIN: 326-460-322

powershell
python

```
PC@LAPTOP-K3A3MUPE MINGW64 /d/MT_API
$ curl -i "http://127.0.0.1:5000/books/abc-1234"
HTTP/1.1 404 NOT FOUND
Server: Werkzeug/3.1.8 Python/3.14.3
Date: Sat, 12 Sep 2026 03:45:29 GMT
Content-Type: application/json
Content-Length: 27
Connection: close

{
  "error": "not found"
}
(.venv)
PC@LAPTOP-K3A3MUPE MINGW64 /d/MT_API
$ curl -i "http://127.0.0.1:5000/items/10"
HTTP/1.1 200 OK
Server: Werkzeug/3.1.8 Python/3.14.3
Date: Sat, 12 Sep 2026 03:46:28 GMT
Content-Type: application/json
Content-Length: 15
Connection: close

{
  "id": 10
}
(.venv)
```

```

PC@LAPTOP-K3A3MUPE MINGW64 /d/MT_API
$ curl -i "http://127.0.0.1:5000/books?limit=5&q=python"
HTTP/1.1 200 OK
Server: Werkzeug/3.1.8 Python/3.14.3
Date: Sat, 12 Sep 2026 03:46:37 GMT
Content-Type: application/json
Content-Length: 80
Connection: close

{
  "items": [
    {
      "id": 2,
      "t": "Python Programming"
    }
  ]
}
(.venv)

```

Bài 5:

```

app.py X
app.py > ...
1 app = Flask(__name__)
2
3 ORDERS = {
4     "1": {"id": "1", "status": "pending"},
5     "2": {"id": "2", "status": "shipped"}
6 }
7 @app.route("/orders/<order_id>", methods=["DELETE"])
8 def delete_order(order_id):
9     order = ORDERS.get(order_id)
10    # 404 - không tìm thấy
11    if order is None:
12        return {"error": "not found"}, 404
13    # 409 business rule
14    if order["status"] in ("shipped", "delivered"):
15        return {"error": "cannot delete"}, 409
16    ORDERS.pop(order_id, None)
17    # 204 success, no body
18    return "", 204
19
20 if __name__ == "__main__":
21     app.run(host="127.0.0.1", port=5000, debug=True)

```

Output Debug Console Terminal Ports

> Terminal

```

(.venv)
PC@LAPTOP-K3A3MUPE MINGW64 /d/MT_API
$ python app.py
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 326-460-322
* Detected change in 'D:\MT_API\app.py', reloading
* Restarting with stat
* Debugger is active!
* Debugger PIN: 326-460-322

```

```
PC@LAPTOP-K3A3MUPE MINGW64 /d/MT_API
$ curl -i -X DELETE "http://127.0.0.1:5000/orders/1"
HTTP/1.1 204 NO CONTENT
Server: Werkzeug/3.1.8 Python/3.14.3
Date: Sat, 12 Sep 2026 03:55:50 GMT
Content-Type: text/html; charset=utf-8
Connection: close

(.venv)
PC@LAPTOP-K3A3MUPE MINGW64 /d/MT_API
$ curl -i -X DELETE "http://127.0.0.1:5000/orders/2"
HTTP/1.1 409 CONFLICT
Server: Werkzeug/3.1.8 Python/3.14.3
Date: Sat, 12 Sep 2026 03:55:58 GMT
Content-Type: application/json
Content-Length: 31
Connection: close

{
  "error": "cannot delete"
}
(.venv)
PC@LAPTOP-K3A3MUPE MINGW64 /d/MT_API
$ curl -i -X DELETE "http://127.0.0.1:5000/orders/999"
HTTP/1.1 404 NOT FOUND
Server: Werkzeug/3.1.8 Python/3.14.3
Date: Sat, 12 Sep 2026 03:56:12 GMT
Content-Type: application/json
Content-Length: 27
Connection: close

{
```

Bài 6:

```
app.py X
app.py > modify_book
23 def create_book():
24     return {"error": "need title+author"}, 400
29     book = {"id": _next, "title": t, "author": a}
30     _next += 1
31     BOOKS.append(book)
32     return jsonify(book), 201, {"Location": f"/books/{book['id']}"}
33 # UPDATE PUT, DELETE DELETE
34 @app.route("/books/<int:bid>", methods=["PUT", "DELETE"])
35 def modify_book(bid):
36     book = find(bid)
37     if not book:
38         return {"error": "not found"}, 404
39     if request.method == "PUT":
40         book.update(request.get_json(silent=True) or {})
41         return jsonify(book), 200
42     BOOKS.remove(book)
43     return "", 204
44 if __name__ == "__main__":
45     app.run(host="127.0.0.1", port=5000, debug=True)
```

Output Debug Console Terminal Ports

> Terminal

```
PC@LAPTOP-K3A3MUPE MINGW64 /d/MT_API
$ source /d/MT_API/.venv/Scripts/activate
(.venv)
PC@LAPTOP-K3A3MUPE MINGW64 /d/MT_API
$ python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 326-460-322
```

```
PC@LAPTOP-K3A3MUPE MINGW64 /d/MT_API
$ curl -i "http://127.0.0.1:5000/books"
HTTP/1.1 200 OK
Server: Werkzeug/3.1.8 Python/3.14.3
Date: Sat, 12 Sep 2026 03:59:15 GMT
Content-Type: application/json
Content-Length: 78
Connection: close

[
  {
    "author": "R. Martin",
    "id": 1,
    "title": "Clean Code"
  }
]
(.venv)
PC@LAPTOP-K3A3MUPE MINGW64 /d/MT_API
$ curl -i -X POST "http://127.0.0.1:5000/books" -H "Content-Type: application/json" -d "{\"title\": \"DDIA\", \"author\": \"Kleppmann\"}"
HTTP/1.1 201 CREATED
Server: Werkzeug/3.1.8 Python/3.14.3
Date: Sat, 12 Sep 2026 03:59:23 GMT
Content-Type: application/json
Content-Length: 58
Location: /books/1
Connection: close

{
  "author": "Kleppmann",
  "id": 1,
  "title": "DDIA"
}
(.venv)
```

```
PC@LAPTOP-K3A3MUPE MINGW64 /d/MT_API
$ curl -i "http://127.0.0.1:5000/books/1"
HTTP/1.1 200 OK
Server: Werkzeug/3.1.8 Python/3.14.3
Date: Sat, 12 Sep 2026 03:59:35 GMT
Content-Type: application/json
Content-Length: 64
Connection: close

{
  "author": "R. Martin",
  "id": 1,
  "title": "Clean Code"
}
(.venv)
PC@LAPTOP-K3A3MUPE MINGW64 /d/MT_API
$ curl -i "http://127.0.0.1:5000/books/99"
HTTP/1.1 404 NOT FOUND
Server: Werkzeug/3.1.8 Python/3.14.3
Date: Sat, 12 Sep 2026 03:59:45 GMT
Content-Type: application/json
Content-Length: 27
Connection: close

{
  "error": "not found"
}
(.venv)
```

```
PC@LAPTOP-K3A3MUPE MINGW64 /d/MT_API
$ curl -i -X PUT "http://127.0.0.1:5000/books/1" -H "Content-Type: application/json" -d "{\"ti
tle\": \"CC 2nd ed.\"}"
HTTP/1.1 200 OK
Server: Werkzeug/3.1.8 Python/3.14.3
Date: Sat, 12 Sep 2026 03:59:53 GMT
Content-Type: application/json
Content-Length: 64
Connection: close

{
  "author": "R. Martin",
  "id": 1,
  "title": "CC 2nd ed."
}
(.venv)
PC@LAPTOP-K3A3MUPE MINGW64 /d/MT_API
$ curl -i -X DELETE "http://127.0.0.1:5000/books/1"
HTTP/1.1 204 NO CONTENT
Server: Werkzeug/3.1.8 Python/3.14.3
Date: Sat, 12 Sep 2026 04:00:41 GMT
Content-Type: text/html; charset=utf-8
Connection: close

(.venv)
PC@LAPTOP-K3A3MUPE MINGW64 /d/MT_API
$ curl -i -X POST "http://127.0.0.1:5000/books" -H "Content-Type: application/json" -d "{}"
HTTP/1.1 400 BAD REQUEST
Server: Werkzeug/3.1.8 Python/3.14.3
Date: Sat, 12 Sep 2026 04:01:09 GMT
Content-Type: application/json
Content-Length: 35
Connection: close

{
  "error": "need title+author"
}
```